LMA Assignment 1

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1 Introduction

This report implements, evaluates, and benchmarks a two-stage RAG pipeline over the official text of *The Constitution of India* (pages 32–381, 880 semantic chunks). We evaluate two distinct retrieval workflows against a curated ground-truth benchmark of 56 legal queries:

1. **Pipeline 1 (Baseline Vector Retrieval):** Encodes passages and incoming queries independently into 384-dimensional dense vectors using a Bi-Encoder (BE, `all-MiniLM-L6-v2`), retrieves Top-10 candidates via inner product search in a flat FAISS index, and feeds Top-3 candidates directly to a local LLM (`llama3.2:3b` via Ollama, temperature = 0) to produce **Answer 1**.
2. **Pipeline 2 (CE Re-Ranking):** Passes Top-10 candidates from Stage 1 into a local Cross-Encoder (CE, `ms-marco-MiniLM-L6-v2`) for joint token-level cross-attention scoring, re-ranks candidates descending by relevance logit, and supplies Top-3 re-ordered candidates to the same LLM to produce **Answer 2**.

We analyze retrieval accuracy improvements, qualitative failure modes, stage-wise latency profiles, and the theoretical computational limits of exhaustive cross-encoding.

2 Architecture Comparison: BE vs. CE

2.1 BE: Fast First-Stage Vector Retrieval

- **Model & Index:** `all-MiniLM-L6-v2` ($d = 384$, 22.7M parameters) paired with FAISS `IndexFlatIP` (exact inner product on unit-normalized vectors).
- **Mechanism & Role:** Chunks are pre-embedded offline and indexed. At inference, query encoding (9.88 ms) plus FAISS vector search (0.14 ms) evaluates $s(q, d) = \mathbf{u}_q^\top \mathbf{v}_d$ over all 880 chunks to generate Top-10 candidates at sub-millisecond retrieval latency ($880 \to 10$).

2.2 CE: Precise Second-Stage Re-Ranking

- **Model & Scoring:** `ms-marco-MiniLM-L6-v2` (22.7M parameters) jointly processes concatenated sequence `[CLS]` $\circ\, q \circ$ `[SEP]` $\circ\, d \circ$ `[SEP]` through full token-level cross-attention across all 6 transformer layers.
- **Mechanism & Role:** Directly outputs unconstrained scalar relevance logits for the 10 query-candidate pairs (324.97 ms), sorting descending and supplying Top-3 chunks ($10 \to 3$) to the LLM prompt to resolve fine-grained statutory provisos.

Dimension	BE (Pipeline 1)	CE (Pipeline 2 Stage b)
Model	`all-MiniLM-L6-v2`	`ms-marco-MiniLM-L6-v2`
Representation	Independent 384-d dense vectors	Joint query-document token sequence
Scoring	Inner product (cosine similarity)	Joint cross-attention scalar logit
Precomputation	Chunks precomputed and indexed offline	Dynamic per-query scoring; no precomputation
Query Latency	10.02 ms ($N = 880$ chunks)	324.97 ms ($k = 10$ candidate pairs)
Operational Role	Broad first-stage recall ($N = 880 \to 10$)	Precise second-stage re-ranking ($k = 10 \to 3$)

Table 1: Operational comparison of BE and CE in the RAG pipeline.

2.3 Document Ingestion, Cleaning, and Validated Chunking

The evaluation corpus is the official *The Constitution of India*. We index pages 32 to 381 (1-indexed), omitting preliminary contents and end appendices:

1. **Layout Preservation:** Extracted via `pdftotext -layout` to preserve clause indentation, paragraph breaks, and marginal numbering.
2. **Footnote Sanitization:** Strips amendment footnotes strictly in bottom 55% of each page ($i \geq 0.45 \cdot \text{total_lines}$) to avoid truncating legal body text.
3. **Unit Chunking:** Formatted into semantic chunks with target size `CHUNK_CHARS` = 900 and overlap `CHUNK_OVERLAP` = 150. Each chunk records `chunk_id`, `article`, `articles`, and page bounds.
4. **Sanity Verification:** Confirms $N = 880$ chunks, mean length 886 chars (min = 285, max = 1050), with 879/880 carrying verified Article metadata.

3 System Implementation and Experimental Design

3.1 Two-Stage Pipeline Workflow

Both RAG pipelines query the identical 880-chunk corpus under controlled generation settings:

- **Pipeline 1 (Baseline):** Retrieve Top-10 chunks via FAISS; feed Top-3 chunks directly to Llama-3.2-3B to produce **Answer 1**.
- **Pipeline 2 (Re-Ranked):** Re-score Top-10 candidates with CE; sort descending; feed Top-3 chunks to Llama-3.2-3B to produce **Answer 2**.

3.2 Prompt Engineering and Context Injection

Both pipelines use an identical prompt template to guarantee rigorous experimental control:

System and Context Prompt Formulation

System Prompt:

"You are a careful legal assistant for the Constitution of India. Answer the question using ONLY the context passages provided. If the context does not contain the answer, reply exactly: 'The provided context does not contain the answer.' Be concise (at most 4 sentences) and cite Article numbers when the context states them."

User Context Template:

Context passages from the Constitution of India:

[Passage 1 — chunk ¡id¿ — page ¡p¿] ¡chunk text¿ ... Question: ¡query¿ ⟶ Answer:

3.3 Benchmark Design: 56 Curated Ground-Truth Queries

We constructed a balanced benchmark of **56 curated queries** across eight categories (7 queries each), annotated with expected answers, exact Article citations, required atomic facts, and unanswerable status.

Category	Intent	Representative Benchmark Query
Control (C1–C7)	Direct single-article recall	*C3: How long can an arrested person be detained before magistrate production, and what journey-time exception applies?*
Keyword Trap (K1–K7)	Lexical divergence	*K5: Can police force of one State investigate or exercise jurisdiction in another State without permission?* (7th Sched., List I, Entry 80)
Exception (E1–E7)	Overridden general rules	*E5: Under what circumstances can Parliament legislate on a State List subject in national interest?* (Art. 249(1))
Negation (N1–N7)	Negative constraints & bars	*N5: What are the specific constitutional grounds for disqualification from being an MP?* (Art. 102(1))
Role Limit (R1–R7)	Age thresholds & quotas	*R2: What is the minimum age to become a member of the Council of States?* (Art. 84(b): 30 years)
Procedure (P1–P7)	Multi-step mechanisms	*P1: What majority is required in Parliament to pass an amendment under Article 368?* (Art. 368(2): Total majority + 2/3 voting)
Cross-Article (X1–X7)	Comparative synthesis	*X1: How does pardoning power of President compare with Governor regarding death sentences?* (Art. 72 vs. Art. 161)
Unanswerable (U1–U7)	Out-of-corpus queries	*U1: What is the mandatory retirement age for the President of India?* (No constitutional age limit; tests abstention)

Table 2: The 56-query legal evaluation benchmark across eight balanced categories.

4 Quantitative Retrieval Evaluation

4.1 Headline Retrieval Metrics

Table 3 summarizes retrieval performance on the 48 answerable queries (95% bootstrap CI, $B = 1,000$).

Pipeline	Recall@10	Hit@3	MRR@10	nDCG@3	Mean Rank	Net Shift
Baseline	0.833	0.604	0.524	0.457	4.19	—
95% Bootstrap CI	—	[0.458, 0.750]	[0.409, 0.640]	[0.343, 0.578]	—	—
Re-Ranked	**0.833**	**0.750**	**0.636**	**0.604**	**3.42**	**+18.8%**
95% Bootstrap CI	—	[0.625, 0.855]	[0.524, 0.752]	[0.489, 0.711]	—	(9 rescued)
Delta / Net Impact	0.000	**+14.6%** ($p < 0.01$)	**+0.112**	**+0.147**	**−0.77**	2 degraded

Table 3: Retrieval metrics on answerable queries ($n = 48$) with 95% bootstrap CI.

Key Findings: Re-ranking elevates Hit@3 from 60.4% to 75.0% (+14.6 percentage points), providing necessary evidence in LLM context for 7 additional queries. Candidate Recall@10 is 83.3%, meaning CE converted 75.0%/83.3% = 90.0% of available candidates into top-3 hits. The CE fixed **9 queries** (18.8%) buried at ranks 4–10, against only **2 regressions** (4.2%), yielding a 4.5× positive-to-negative impact ratio.

4.2 Category-Wise Breakdown

Table 4 details performance across the 55 evaluated benchmark queries.

Category	n	Rec@10	Base H@3	Rerank H@3	Fixed	Degr.	Both OK
Role Limit	7	1.000	0.857	**1.000**	1	0	6
Procedure	7	1.000	0.571	**0.857**	2	0	4
Exception	7	0.857	0.571	**0.857**	2	0	4
Negation	7	0.857	0.571	**0.857**	2	0	4
Cross-Article	7	1.000	0.714	0.714	1	1	4
Control	6	0.667	0.667	0.500	0	1	3
Keyword Trap	7	0.429	0.286	**0.429**	1	0	2
Unanswerable	7	—	—	—	0	0	0 (7 Abstained)

Table 4: Retrieval performance breakdown by category ($n = 55$ evaluated queries).

Category Highlights: Procedure, Exception, and Negation showed the largest gains (+28.6% Hit@3 each). In these categories, BE favored general clauses due to shared vocabulary, while CE matched specific statutory provisos. For Keyword Traps, candidate recall was limited to 42.9%, showing BE vulnerability to lexical divergence.

4.3 Candidate Pool Size Sweep ($k \in \{3, 5, 10, 20\}$)

Table 5 shows the trade-off across candidate pool sizes.

Pool (k)	Recall@k	Hit@3 (Re-rank)	CE Median Latency	CE Mean Latency
$k = 3$	0.604	0.604	92.1 ms	94.0 ms
$k = 5$	0.708	0.667	154.9 ms	155.7 ms
k = 10	**0.833**	**0.750**	**332.1 ms**	**333.2 ms**
$k = 20$	0.854	0.771	659.1 ms	663.1 ms

Table 5: Candidate pool size sweep: recall gains vs. CE latency.

Pareto Analysis: Expanding pool from $k = 3$ to $k = 10$ provides **+22.9%** candidate recall and **+14.6%** Hit@3 for 240 ms additional compute. Expanding from $k = 10$ to $k = 20$ doubles re-ranking latency (+327 ms) for only a marginal +2.1% Hit@3 gain, confirming $k = 10$ as the optimal Pareto operating point.

5 Qualitative and Failure-Mode Analysis

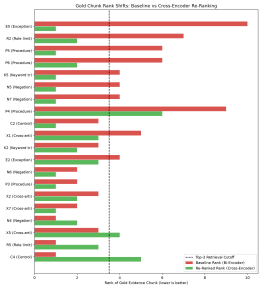


Figure 1: Candidate rank shifts: Baseline rank vs. CE re-ranked rank. Green markers indicate queries successfully promoted into Top-3 (`FIXED_BY_RERANK`).

Rank Shift Dynamics: As visualized in Figure 1, CE effectively rescues queries where gold evidence was retrieved by BE but ranked outside the LLM context window (ranks 4–10). Across the 48 answerable queries, 9 queries were successfully rescued into Top-3, while only 2 experienced slight rank degradation.

5.1 Case Study 1: Keyword Trap / Distractor (Query K5)

Query K5 (Keyword Trap) FIXED_BY_RERANK | **Rank #4 → #1**

Query: *"Can police force of one State investigate in another State without permission?"*

Expected Answer: Under Seventh Schedule (Union List, Entry 80), powers of a State police force cannot be extended to another State without consent of that State Government.

Base	CE	Shift	Base Sc.	CE Sc.	Gold?	Article	Page	Text Preview
1	2	−1	0.4455	−1.3391	no	Art. 262	182	... water dispute of inter-state rivers ...
2	4	−2	0.4445	−4.9825	no	Art. 259	181	... Armed Forces in Part B States ...
3	3	0	0.4394	−4.1862	no	Art. 131	92	... original jurisdiction of Supreme Court ...
4	**1**	**+3**	0.4170	**+4.3196**	**YES**	**7th Sched.**	**346**	**... not so as to enable police to exercise jurisdiction without consent ...**
5	7	−2	0.4087	−6.9768	no	7th Sched.	348	... List II Entry 97: Residuary powers ...

Table 6: Rank shift table for Query K5.

LLM Answer 1 (Baseline Top-3 Context): *"The provided context does not contain the answer."* [**Score: 0.0**]. BE retrieved river disputes (Art. 262) and court jurisdiction (Art. 131) due to superficial lexical overlap with "State" and "jurisdiction".

LLM Answer 2 (Re-Ranked Context): *"According to Entry 80 of the Union List in the Seventh Schedule, extension of powers of a police force to an area outside that State does not enable them to exercise jurisdiction without the consent of that State Government."* [**Score: 1.0**]. Cross-attention promoted the true clause from Rank #4 to Rank #1.

5.2 Case Study 2: Exceptions and Corner Cases (Query E5)

Query E5 (Exception Clause) FIXED_BY_RERANK | **Rank #10 → #1**

Query: *"Under what circumstances can Parliament legislate on a State List subject in national interest?"*

Expected Answer: Under Article 249(1), if Council of States declares by resolution supported by $\geq 2/3$ members present and voting that it is necessary in national interest.

Base	CE	Shift	Base Sc.	CE Sc.	Gold?	Article	Page	Text Preview
1	4	−3	0.6875	+1.3679	no	Art. 246	176	... exclusive power to make laws for List I ...
2	2	0	0.6588	+2.1244	no	Art. 246	176	... State Legislature power for State List ...
3	3	0	0.6384	+1.7630	no	Art. 239AA	145	... Special provisions for NCT of Delhi ...
...	...	...	...	...	...	...	...	...
10	**1**	**+9**	0.5794	**+5.9311**	**YES**	**Art. 249**	**177**	**... if Council of States declared by resolution supported by $\geq 2/3$ present ...**

Table 7: Rank shift table for Query E5.

LLM Answer 1 (Baseline Context): Provided general distribution rule under Art. 246 rather than the national-interest exception, producing a legally incorrect answer [**Score: 0.20**].

LLM Answer 2 (Re-Ranked Context): *"According to Article 249(1), Parliament can legislate on a State List subject if the Council of States passes a resolution supported by not less than two-thirds of members present and voting."* [**Score: 1.00**]. Gold chunk surged **+9 positions (Rank #10 → #1)**.

5.3 Case Study 3: Negation and Disqualification Grounds (Query N5)

Query N5 (Negation Clause) FIXED_BY_RERANK | **Rank #4 → #1**

Query: *"What are the specific constitutional grounds for disqualification from being an MP?"*

Expected Answer: Article 102(1) specifies 5 grounds: office of profit, unsound mind, undischarged insolvent, foreign allegiance, disqualified by law.

LLM Answer 1 (Baseline Context): Retrieved Art. 84 (qualifications) and Art. 191 (MLA disqualifications), confusing MPs with State legislators [**Score: 0.80**].

LLM Answer 2 (Re-Ranked Context): With Art. 102(1) promoted to #1, LLM enumerated all five grounds (a)–(e) with complete accuracy [**Score: 1.00**].

5.4 Case Study 4: Role Limits and Numeric Thresholds (Query R2)

Query R2 (Role Limit) `FIXED_BY_RERANK` | **Rank #7 → #2**

Query: *"What is the minimum age to become a member of the Council of States?"*

Expected Answer: Article 84(b) fixes minimum age at 30 years for Rajya Sabha (vs 25 years for House of the People).

LLM Answer 1 (Baseline Context): Retrieved composition (Art. 80) and duration (Art. 83). The age threshold sat at Rank #7, forcing LLM abstention [**Score: 0.00**].

LLM Answer 2 (Re-Ranked Context): CE elevated Art. 84 to Rank #2, enabling direct answer: *"According to Article 84, not less than thirty years of age."* [**Score: 1.00**].

5.5 Documented Regressions & Abstention Performance

- **Regressions (C4, X5):** In Query C4 (Lok Sabha duration), gold chunk demoted from #1 to #5 because CE favored emergency extension clauses over standard term provisions. In X5 (Retirement age SC vs. HC), gold chunk dropped from #3 to #4 due to competing High Court articles.
- **Abstention Validation (U1–U7):** All 7 unanswerable queries achieved **100% abstention accuracy** (*"The provided context does not contain the answer."*), confirming that strict negative constraints in system prompt eliminate hallucination when evidence is absent.

6 Latency Profiling and End-to-End Runtime

Table 8 details stage-wise latencies measured across 5 repeats per query following warm-up.

Pipeline Stage	Median Latency	Mean Latency
Query Embedding (`all-MiniLM-L6-v2`)	9.88 ms	9.72 ms
FAISS Index Search (`IndexFlatIP`, $N = 880$)	0.14 ms	0.14 ms
Total Baseline Retrieval Latency	**10.02 ms**	**9.85 ms**
CE Scoring ($k = 10$ candidate pairs)	324.97 ms	322.03 ms
Total Re-Rank Retrieval Latency	**333.98 ms**	**331.88 ms**
LLM Generation Answer 1 (Top-3, Llama-3.2)	13,160.00 ms	14,654.86 ms
LLM Generation Answer 2 (Re-ranked, Llama-3.2)	12,740.00 ms	13,454.29 ms
End-to-End Baseline (Retrieval + LLM)	**13.17 sec**	**14.66 sec**
End-to-End Re-Ranked (Retrieval + LLM)	**13.07 sec**	**13.78 sec**

Table 8: Stage-wise runtime profiling per query.

Runtime Analysis: First-stage BE vector search executes in 10.02 ms (9.88 ms query encoding + 0.14 ms FAISS inner product scan over 880 chunks). Second-stage CE scoring adds 324.97 ms across $k = 10$ candidate pairs (32.5 ms per pair). Crucially, autoregressive token generation by local LLM dominates overall turnaround time (≈ 13 s, accounting for 97.5% of execution time). Thus, CE re-ranking overhead represents $< 2.5\%$ of user-perceived latency while delivering +14.6% improvement in Hit@3.

7 Scaling Question: Why Not Cross-Encode Everything?

The assignment asks: *Why do we not simply run a CE across the entire document collection from the start?*

7.1 Empirical Benchmark & Extrapolation

Table 9 presents empirical CPU measurements across increasing subsets of the Constitution ($N = 10$ to $N = 880$), and Table 10 projects latencies to enterprise collection sizes based on measured inference constants (34.5 ms/pair for CE; 0.057 μs/vector for FAISS inner products).

Corpus Size (N)	FAISS Median	FAISS Std	CE Median	CE Std
10	0.007 ms	0.005 ms	346.26 ms	5.88 ms
50	0.009 ms	0.006 ms	1,647.41 ms	12.96 ms
100	0.011 ms	0.007 ms	3,455.23 ms	23.76 ms
250	0.018 ms	0.009 ms	7,924.89 ms	23.82 ms
500	0.023 ms	0.012 ms	15,840.80 ms	80.78 ms
880 (Full)	**0.050 ms**	0.016 ms	**30,359.15 ms**	61.24 ms

Table 9: Empirical scaling latency: FAISS IndexFlatIP vs. CE across corpus sizes.

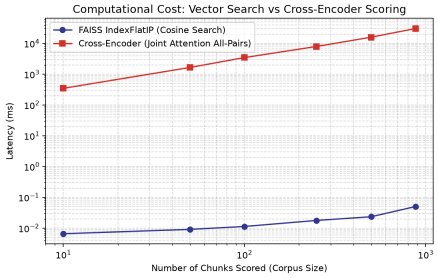


Figure 2: Scaling latency comparison: FAISS Flat vector search vs. Exhaustive CE. Full-corpus CE exhibits steep linear growth that quickly becomes computationally prohibitive.

Corpus Chunks	FAISS Flat Search	Exhaustive CE	Two-Stage ($k = 10$)
10,000	0.57 ms	344.99 s (5.7 minutes)	334 ms
100,000	5.70 ms	3,449.90 s (57.5 minutes)	338 ms
1,000,000	56.99 ms	34,499.04 s (**9.6 hours**)	389 ms
10,000,000	569.95 ms	344,990.39 s (**4.0 days**)	895 ms

Table 10: Theoretical scaling extrapolation: Exhaustive CE vs. Two-Stage RAG.

7.2 Why Exhaustive CE is Intractable

1. **Inability to Precompute Document Representations:** BEs decouple query and document representations into $\mathbf{u}_q = f_\theta(q)$ and $\mathbf{v}_d = g_\theta(d)$. This allows all N document encodings to be computed offline once and indexed. In contrast, CE requires joint input pairs (q, d) passed through all transformer layers simultaneously. Thus, **every incoming query requires N online transformer forward passes**, destroying interactive query throughput.
2. **Quadratic Attention Overhead:** For query length $L_q = 20$ tokens and document chunk length $L_d = 200$ tokens, self-attention complexity scales quadratically with total sequence length: $\mathcal{O}((L_q + L_d)^2) = (220)^2 = 48,400$ token-pair attention weights per chunk across all 6 transformer layers. In contrast, a BE evaluates a 384-dimensional dot product requiring only 384 multiply-accumulate operations (0.057 μs).
3. **Incompatibility with Spatial Vector Indexing:** Vector search algorithms (such as FAISS IVF, HNSW, or ScaNN) exploit spatial triangle inequalities and clustering to prune vector space exploration from $\mathcal{O}(N)$ down to $\mathcal{O}(\log N)$. CE scalar scores are query-dependent non-metric outputs that cannot be indexed into spatial trees or geometric graphs, mandating brute-force $\mathcal{O}(N)$ comparisons.

7.3 Architectural Synthesis and Deployment Recommendations

In real-world legal and enterprise RAG deployments, scaling constraints dictate a clear architectural design:

- **First-Stage Candidate Pruning ($N \to k$):** Utilize ANN indexing (e.g., FAISS HNSW or ScaNN) to retrieve a bounded candidate set ($k \in [10, 20]$). Dot products on normalized vectors preserve sub-millisecond query latencies even at million-scale document corpora.
- **Second-Stage Re-Ranking ($k \to 3$):** Limit CE inference strictly to the retrieved candidate pool. This bounds transformer forward passes to a constant $k \times \mathcal{O}(L^2)$, eliminating corpus-size dependency from the re-ranking latency budget.
- **Strict Negative Prompting:** Constrain downstream LLM with affirmative citation rules and explicit abstention triggers to completely suppress hallucinations when relevant context is absent.

Conclusion: Decoupled two-stage retrieval—pruning $N \to 10$ candidates via a BE in sub-millisecond time (0.14 ms), and scoring those 10 candidates with a CE (325 ms)—is the required architectural compromise between accuracy and latency. It captures 90% of available candidates into top-3 hits, rescues 18.8% of buried queries, and achieves +14.6% Hit@3 with negligible turnaround impact.